

(*) On every Riemannian manifold, we will always use the Levi-Civita connection if there is no further comment.

1 Gauss Theorema Egregium

1.1 Algebraic Curvature Tensor and Kulkarni-Nomizu Product

Recall that we have introduced the symmetries of Rm . Now, we extract the key properties of the symmetries of Rm and give the following definition.

Algebraic Curvature Tensor

Definition 1.1. : Suppose V is an n -dimensional vector space. Let $\mathcal{R}(V^*) \subseteq T^4(V^*)$ denote the vector space of all covariant 4-tensors T on V that have the following symmetries:

- (1) $T(w, x, y, z) = -T(x, w, y, z)$;
- (2) $T(w, x, y, z) = -T(w, x, z, y)$;
- (3) $T(w, x, y, z) = T(y, z, w, x)$;
- (4) $T(w, x, y, z) + T(x, y, w, z) + T(y, w, x, z) = 0$.

An element of $\mathcal{R}(V^*)$ is called an **algebraic curvature tensor** on V .

There is a natural way to construct an algebraic curvature tensor out of two symmetric 2-tensors, which is given as follows.

For any $h, k \in \Sigma^2(V^*)$, we define a covariant 4-tensor $h \lrcorner k$, called **Kulkarni-Nomizu product** of h and k , which is given by

$$h \lrcorner k(w, x, y, z) := h(w, z)k(x, y) + h(x, y)k(w, z) - h(w, y)k(x, z) - h(x, z)k(w, y)$$

it's clear to see that $h \lrcorner k$ satisfies the symmetries in the definition above, so $h \lrcorner k \in \mathcal{R}(V^*)$, hence we have a map $\lrcorner : \Sigma^2(V^*) \times \Sigma^2(V^*) \rightarrow \mathcal{R}(V^*)$.

Properties of Kulkarni-Nomizu Product

Proposition 1.1. : Let the assumption be as in the discussion above, let g be a scalar product in V , then we have

- (1) $h \lrcorner k = k \lrcorner h$;
- (2) $\text{tr}_g(h \lrcorner g) = (n - 2)h + (\text{tr}_g h)g$;
- (3) $\text{tr}_g(g \lrcorner g) = 2(n - 1)g$.

Proof: (1) is by the definition and an easy check; (2) is by the calculation in components once we take a basis; (3) is a special case of (2). $\square$

1.2 Hypersurfaces

Recall that we have defined the scalar symmetric bilinear form $\Pi_N : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow C^\infty(M)$ to be $\Pi_N(X, Y) := \langle N, \Pi(X, Y) \rangle$, where $N \in \Gamma(NM)$ is a smooth normal vector field. Now, let M be

a **hypersurface** of $\widetilde{M}$, i.e., M is an embedded submanifold of $\widetilde{M}$ with codimension 1. We define the **scalar second fundamental form of M** to be the covariant 2-tensor field $h \in \Gamma(\Sigma^2 T^*M)$, which is given by

$$h(X, Y) := \Pi_N(X, Y) = \langle N, \Pi(X, Y) \rangle$$

By the Gauss formula, for $X, Y \in \mathfrak{X}(M)$, we have $\widetilde{\nabla}_X Y = \nabla_X Y + \Pi(X, Y)$ and $\nabla_X Y \in \mathfrak{X}(M)$, so we know that $h(X, Y) = \langle N, \widetilde{\nabla}_X Y \rangle$. More over, if we choose N as a unit normal vector field on M , we can also gain $\Pi(X, Y) = h(X, Y)N$.

Recall that for a unit normal vector field, we have defined the self-adjoint Weingarten map $W_N : \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$, which is given by $\langle W_N(X), Y \rangle = \Pi_N(X, Y) = \langle N, \Pi(X, Y) \rangle$. Here, in the case of a hypersurface, we denote $s := W_N$ and call it **shape operator of M** , hence it is given by

$$\langle s(X), Y \rangle = h(X, Y) = \Pi_N(X, Y) = \langle N, \Pi(X, Y) \rangle \quad \text{for } \forall X, Y \in \mathfrak{X}(M)$$

and more over, $\langle s(X), Y \rangle = \langle X, s(Y) \rangle$.

Fundamental Equation for a Hypersurface

Theorem 1.1. : Let (M, g) be a hypersurface of $(\widetilde{M}, \widetilde{g})$, and N is a smooth unit normal vector field along M , then we have

(1) (**Gauss Formula for a Hupersurface**) If $X, Y \in \mathfrak{X}(M)$ are extended to an open subset of $\widetilde{M}$, then

$$\widetilde{\nabla}_X Y = \nabla_X Y + h(X, Y)N$$

(2) (**Gauss Formula for a Curve in a Hypersurface**) For a smooth curve $\gamma : I \rightarrow M$ and $X : I \rightarrow TM \in \mathfrak{X}(\gamma)$, then

$$\widetilde{D}_t X = D_t X + h(\gamma', X)N$$

(3) (**Weingarten Equation for a Hypersurface**) For $\forall X \in \mathfrak{X}(M)$, we have

$$\widetilde{\nabla}_X N = -s(X)$$

(4) (**Gauss Equation for a Hypersurface**) For $\forall W, X, Y, Z \in \mathfrak{X}(M)$,

$$\widetilde{Rm}(W, X, Y, Z) = Rm(W, X, Y, Z) - \frac{1}{2}(h \lrcorner h)(W, X, Y, Z)$$

(5) (**Codazzi Equation for a Hypersurface**) For $\forall W, X, Y, Z \in \mathfrak{X}(M)$, we have

$$\widetilde{Rm}(W, X, Y, N) = (Dh)(Y, W, X)$$

Proof: (1), (2) and (4) is gained by the fact that $h(X, Y)N = \Pi(X, Y)$ and the definition of Kulkarni-Nomizu product;

To prove (3), note that in the general case we have $(\widetilde{\nabla}_X N)^\top = -W_N(X)$, so $(\widetilde{\nabla}_X N)^\top = -s(X)$. Since $\frac{1}{2}(\widetilde{\nabla}_X \langle N, N \rangle) = \frac{1}{2}(X \langle N, N \rangle) = \langle \widetilde{\nabla}_X N, N \rangle \equiv 0$, hence $\widetilde{\nabla}_X N \in \mathfrak{X}(M)$, so $(\widetilde{\nabla}_X N)^\top = \widetilde{\nabla}_X N = -s(X)$;

Now we prove (5), note that N is a unit normal vector field, which implies that

$$X\langle N, N \rangle_{\tilde{g}} = 2\langle \tilde{\nabla}_X N, N \rangle_{\tilde{g}} = 2\langle (\tilde{\nabla}_X N)^\perp, N \rangle_{\tilde{g}} = 2\langle \nabla_X^\perp N, N \rangle_{\tilde{g}} \equiv 0$$

where $\nabla^\perp$ is the normal connection. By the definition of the connection ∇^F , we have

$$(\nabla_W^F \Pi)(X, Y) = \nabla_W^\perp(\Pi(X, Y)) - \Pi(\nabla_W X, Y) - \Pi(X, \nabla_W Y)$$

since $h(X, Y)N = \Pi(X, Y)$, so we get

$$\begin{aligned} (\nabla_W^F \Pi)(X, Y) &= \nabla_W^\perp(h(X, Y)N) - \Pi(\nabla_W X, Y) - \Pi(X, \nabla_W Y) \\ &= (W(h(X, Y)) - h(\nabla_W X, Y) - h(X, \nabla_W Y))N = (\nabla_W h(X, Y))N \end{aligned}$$

recall that the Codazzi equation in the general case is

$$(\tilde{R}(W, X)Y)^\perp = (\nabla_W^F \Pi)(X, Y) - (\nabla_X^F \Pi)(W, Y)$$

since $h \in \Sigma^2(T^*M)$, hence ∇h is symmetric in its first two indices, from which we obtain

$$\begin{aligned} \widetilde{Rm}(W, X, Y, Z) &= \langle \tilde{R}(W, X)Y, N \rangle = \langle (\tilde{R}(W, X)Y)^\perp, N \rangle = (\nabla_W h)(X, Y) - (\nabla_X h)(W, Y) \\ &= (\nabla h)(X, Y, W) - (\nabla h)(W, Y, X) = (\nabla h)(Y, X, W) - (\nabla h)(Y, W, X) = (Dh)(Y, W, X) \end{aligned}$$

which is the desired equation. $\square$

Specially, when $\widetilde{M} = \mathbb{R}^{n+1}$ is the Euclidan space, the hypersurface $M \subseteq \mathbb{R}^{n+1}$ is endowed with the metirc $g = \iota^* \bar{g}$, where $\bar{g}$ is the Euclidean metirc on $\mathbb{R}^{n+1}$. In this case, we have known that $\overline{Rm} \equiv 0$. Therefore, by the theorem above, we get

$$\frac{1}{2}h \lrcorner h = Rm \tag{1}$$

$$Dh = 0 \tag{2}$$

in terms of components, this is

$$h_{il}h_{jk} - h_{ik}h_{jl} = R_{ijkl} \tag{3}$$

$$h_{ij;k} - h_{ik;j} = 0 \tag{4}$$

1.3 Principal Curvatures, Gaussian Curvatures and Mean Curvatures

Let M^n be a hypersurface of $\widetilde{M}^{n+1}$, then $s : \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ is a self-adjoint linear map w.r.t. g . By the theory of linear algebra, we can know that for each $p \in M$, there is an orthonormal basis $(b_1, b_2, \dots, b_n)$ for $T_p M$ such that $s(b_i) = \kappa_i b_i$ and $\kappa_i \in \mathbb{R}$ for $i = 1, 2, \dots, n$ (see [3] P389-P395). These eigenvalues of s at $p \in M$ are called **principle curvatures** of M at p and the corresponding eigenspaces are called **principle directions**. We define the **Gaussian curvature** to be

Gaussian Curvature

$$K := \det(s) = \prod_{i=1}^n \kappa_i$$

and define the **mean curvature** to be

Mean Curvature

$$H := \frac{1}{n} \text{tr}(s) = \frac{1}{n} \sum_{i=1}^n \kappa_i$$

1.4 Gaussian Curvature of a Surface is Intrinsic

Let M be a surface, i.e., M is a 2-dimensional embedded submanifold of $\mathbb{R}^3$, there is an amazing discovery made by Gauss, which asserts that the Gaussian curvature of a surface M is actually intrinsic invariant of the Riemannian manifold (M, g) .

Gauss's Theorema Egregium

Theorem 1.2. : *Let (M, g) be an embedded 2-dimensional Riemannian manifold of $\mathbb{R}^3$. For each $p \in M$, the Gaussian curvature of M at p is precisely equal to one-half of the scalar curvature of g at p , i.e.,*

$$K(p) = \frac{1}{2} S(p)$$

which implies that the Gauss curvature is a local isometry of (M, g) .

Proof: For any $p \in M$, choose an orthonormal basis (b_1, b_2) of $T_p M$, hence the matrix $(g_{ij}) = I_2$. Then by the definition of s and h , it can be seen that the representation matrix of s is $(s_j^i) = (h_{ij})$ since $h_{ij} = h_{ji}$ because h is symmetric. Thus we know that

$$K(p) = \det(s_j^i) = \det(h_{ij})$$

by the equality (3) in the previous subsection, we get

$$Rm_p(b_1, b_2, b_2, b_1) = h_{11}h_{22} - h_{12}h_{21} = \det(h_{ij}) = \det(s_j^i) = K(p)$$

We can show that $Rm = \frac{1}{4} Sg \lrcorner g$ (see [1] P215-P216 for more details), hence by the definition of Kulkarni-Nomizu product, we obtain

$$Rm_p(b_1, b_2, b_2, b_1) = \frac{1}{4} S(p) (2g_{11}g_{22} - 2g_{12}g_{21}) = \frac{1}{2} S(p)$$

thus

$$K(p) = \frac{1}{2} S(p)$$

which is the conclusion. $\square$

Remark: For an abstract 2-dimensional Riemannian manifold, which is not required to be embedded in $\mathbb{R}^3$, we define its Gaussian curvature to be $K := \frac{1}{2} S$, where S is the scalar curvature.

2 Sectional Curvature

Let M be a Riemannian n -dimensional manifold with $n \geq 2$, recall that for the exponential map $\exp : TM \rightarrow M$ on M satisfies $(d\exp_p)_0 : T_0 T_p M \cong T_p M \rightarrow T_p M$ is precisely the identity

map $\text{Id}_{T_p M}$, hence by the inverse function theorem on manifolds, there exists a neighborhood V of $0 \in T_p M$ and a neighborhood U of $p \in M$ such that $\exp_p|_V$ is a diffeomorphism onto U . Now, let $\Pi \subseteq T_p M$ be a 2-dimensional linear subspace of $T_p M$. Since $\Pi \cap V$ is an embedded 2-dimensional submanifold of V , hence $S_\Pi := \exp_p(\Pi \cap V)$ is an embedded 2-dimensional submanifold of $U \subseteq M$ containing p , which is called the **plane section determined by Π** .

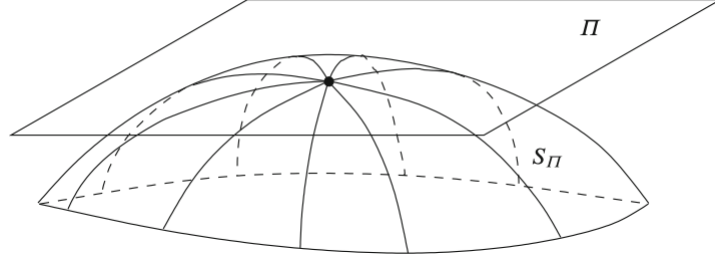


图 1: A plane section

Sectional Curvature

Definition 2.1. : Let Π be as in the discussion above, we define the **sectional curvature** of Π , which is denoted by $\sec(\Pi)$, to be the Gaussian curvature at p of the surface S_Π w.r.t. the metric induced from the embedding $S_\Pi \subseteq M$, i.e.,

$$\sec(\Pi) := K_{S(\Pi)}(p)$$

in particular, if $v, w \in T_p M$ is a basis of $T_p(S_\Pi) = \Pi$, then we denote $\sec(v, w) = \sec(\Pi)$.

Formulas for Sectional Curvature

Proposition 2.1. : Let (M, g) be a Riemannian manifold and $p \in M$. If $v, w \in T_p M$ are linearly independent vectors in $T_p M$, then the sectional curvature of the plane spanned by v, w is given by

$$\sec(v, w) = \frac{Rm_p(v, w, w, v)}{|v \wedge w|^2}$$

where $|v \wedge w| := \sqrt{|v|^2|w|^2 - \langle v, w \rangle^2}$.

Proof: Let $\Pi := \text{span}(v, w)$, let $\widehat{g}$ denote the induced metric on S_Π and its associated curvature tensor is denoted by $\widehat{Rm}$ and so on.

Firstly, we show that the second fundamental form of S_Π in M vanishes at p . For $\forall z \in \Pi$, let $\gamma = \gamma_z$ be the g -geodesic with $\gamma'(0) = z$, whose image lies in S_Π for t sufficiently near 0. Apply the Gauss formula along curves, we gain

$$D_t \gamma' = \widehat{D}_t \gamma' + \Pi(\gamma', \gamma') = 0$$

since $\widehat{D}_t \gamma'$ and $\Pi(\gamma', \gamma')$ are orthogonal to each other, hence $\widehat{D}_t \gamma' = \Pi(\gamma', \gamma') = 0$. Evaluating at $t = 0$ gives $\Pi(z, z) = 0$, so $\Pi = 0$ at p since z is arbitrary. So by the Gauss equation, we get

$$Rm_p(u, v, w, x) = \widehat{Rm}_p(u, v, w, x)$$

for any $u, v, w, x \in \Pi = T_p S_\Pi$. Choose an orthonormal basis (b_1, b_2) for Π , we have

$$\widehat{K}(p) = \frac{1}{2}\widehat{S}(p) = \frac{1}{2} \sum_{i,j=1}^2 \widehat{Rm}_p(b_i, b_j, b_j, b_i) \quad \text{note that } \widehat{Rm}(b_i, b_i, b_i, b_i) = 0 \text{ for } i = 1, 2$$

this equality is obtained from the quality (3) in section 1.2, just as in the proof of the Gauss Theorema Egregium. Hence we have

$$\widehat{K}(p) = \frac{1}{2}\widehat{Rm}_p(b_1, b_2, b_2, b_1) + \frac{1}{2}\widehat{Rm}_p(b_2, b_1, b_1, b_2) = \widehat{Rm}_p(b_1, b_2, b_2, b_1) = Rm_p(b_1, b_2, b_2, b_1)$$

where the second equality is by the symmetry of $\widehat{Rm}$.

Now for any arbitrary linearly independent vectors (v, w) , by the Gram-Schmidt algorithm, we can gain orthonormal basis (b_1, b_2) such that $\text{span}(v, w) = \text{span}(b_1, b_2)$, where

$$b_1 = \frac{v}{|v|}$$

$$b_2 = \frac{w - \langle w, b_1 \rangle b_1}{|w - \langle w, b_1 \rangle b_1|}$$

so we get

$$\begin{aligned} \widehat{K}(p) &= Rm_p(b_1, b_2, b_2, b_1) = Rm_p\left(\frac{v}{|v|}, \frac{w - \langle w, b_1 \rangle b_1}{|w - \langle w, b_1 \rangle b_1|}, \frac{w - \langle w, b_1 \rangle b_1}{|w - \langle w, b_1 \rangle b_1|}, \frac{v}{|v|}\right) \\ &= \frac{Rm_p(v, w, w, v)}{|v \wedge w|^2} \end{aligned}$$

which is the desired equation. $\square$

The next proposition will show that the sectional curvatures completely determine the curvature tensor.

Formulas for Sectional Curvature

Proposition 2.2. : *Let V be a finite-dimensional vector space with an inner product. Suppose $R_1, R_2 \in \mathcal{R}(V^*)$, if for any pair of linearly independent vectors $v, w \in V$,*

$$\frac{R_1(v, w, w, v)}{|v \wedge w|^2} = \frac{R_2(v, w, w, v)}{|v \wedge w|^2}$$

then $R_1 = R_2$.

Proof: Let $D := R_1 - R_2$, hence $D \in \mathcal{R}(V^*)$ and thus we can know from the symmetry of D that $D(v, w, w, v) = 0$ for any $v, w \in V$ even if they are not linearly independent.

For any $v, w, x \in V$, we have

$$D(v+w, x, x, v+w) = D(v, x, x, v) + D(v, x, x, w) + D(w, x, x, v) + D(w, x, x, w) = 2D(v, x, x, w) = 0$$

so it follows that

$$D(v, x+u, x+u, w) = D(v, x, x, w) + D(v, x, u, w) + D(v, u, x, w) + D(v, u, u, w)$$

$$= D(v, x, u, w) + D(v, u, x, w) = 0$$

so by the symmetry of D , we have

$$D(v, w, x, u) + D(w, x, v, u) + D(x, v, w, u) = D(v, w, x, u) - D(w, v, x, u) - D(v, x, w, u) = 3D(v, w, x, u) = 0$$

Since v, w, x, u are all arbitrary, thus we get $D \equiv 0$ and therefore $R_1 = R_2$. $\square$

Geometric Interpretation of Ricci Curvature and Scalar Curvature

Proposition 2.3. : Let (M, g) be an n -dimensional Riemannian manifold and $p \in M$, then

(1) For any unit vector $v \in T_p M$, the Ricci curvature at p is given by

$$\text{Ric}_p(v, v) = \sum_{k=2}^n \sec(v, b_k)$$

where $(v, b_2, \dots, b_n)$ is an orthonormal basis of $T_p M$.

(2) The scalar curvature at p is given by

$$S(p) = \sum_{j \neq k} \sec(b_j, b_k)$$

where $(b_1, b_2, \dots, b_n)$ is any orthonormal basis of $T_p M$.

Proof: (1) Since $v = b_1$, so by the definition of Ricci curvature, we have

$$\text{Ric}_p(v, v) = R_{11}(p) = R_{k11}^k(p) = \sum_{k=2}^n Rm_p(b_k, b_1, b_1, b_k) = \sum_{k=2}^n \sec(b_1, b_k)$$

here note that $Rm_p(b_1, b_1, b_1, b_1) = 0$.

(2) Let $(b_1, \dots, b_n)$ be an orthonormal basis of $T_p M$, by the definition of scalar curvature, we have

$$S(p) = R_j^j(p) = \sum_{j=1}^n \text{Ric}_p(b_j, b_j) = \sum_{j,k=1}^n Rm_p(b_k, b_j, b_j, b_k) = \sum_{j \neq k} \sec(b_j, b_k)$$

by the symmetry of Rm . $\square$

Remark: From this proposition, we can know that if a Riemannian manifold (M, g) has positive (or negative, constant) sectional curvature, then so are its Ricci curvature and scalar curvature.

参考文献

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